Bring Your Own Story: Generating Interactive Visual Novels with AI

Zifan Ye*, Menghan Zhang†, Pratch Suntichaikul‡, Satoru Goto§, Oki Nakamura¶, Ruck Thawonmas‖
*†‡Graduate School of Information Science and Engineering, *Ritsumeikan University*, Ibaraki, Japan
§College of Business Administration, *Ritsumeikan University*, Ibaraki, Japan
¶Institute for General Education, *Ritsumeikan University*, Kyoto, Japan
‖College of Information Science and Engineering, *Ritsumeikan University*, Ibaraki, Japan
{*gr0734sh,†gr0695sp,‡gr0665kx,}@ed.ritsumei.ac.jp, {§goto-s,¶onr14081}@fc.ritsumei.ac.jp, ‖ruck@is.ritsumei.ac.jp

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***Abstract*—This paper presents an automated visual novel (VN) generation system that transforms plain text stories into interactive, branching narrative experiences using the Unity game engine. Unlike prior methods that rely on pre-authored decision points, our system dynamically parses a given story file, segments the narrative into scenes, and injects context-sensitive player choices at runtime. These choices trigger novel story continuations generated by GPT-4o, enabling emergent multi-path storytelling. This end-to-end pipeline, from raw text to full VN rendering, offers a lightweight and scalable tool for interactive narrative prototyping, particularly useful for authors and indie developers. Our contributions include a story segmentation and option generation mechanism, real-time LLM integration, and a Unity-based rendering framework. We conclude by discussing use cases, limitations, and opportunities for future research in generative storytelling and game design.**

***Index Terms*—LLMs (large language models), visual novel, unity, narrative generation**

I. Introduction

Recent progress in large language models (LLMs) has significantly advanced the capabilities of generative storytelling systems, which are essential for visual novels (VNs) [1] —a genre that combines prose fiction with interactivity. Yet despite these capabilities, challenges remain when applying LLMs to generate full-length, interactive stories across multiple choices. Visual novels are a unique form of digital storytelling that combine static visuals, branching narratives, and text-based interaction. Traditionally, VN creation requires substantial manual effort: authors must predefine scenes, craft branching options, and coordinate script and asset implementation in an engine such as Unity or Ren'Py. This process is time-consuming, rigid, and limits rapid prototyping or dynamic content adaptation.

In this work, we propose a system that automates visual novel content generation directly from a plain text story file. Given a narrative in natural language, our system:

1) Automatically segments the story into scenes based on dialogue and paragraph structure,
2) Inserts AI-generated choice points dynamically (not pre-authored),
3) Generates novel story branches using LLM-based completion,
4) Renders the entire experience in Unity, including character dialogue, branching logic, and transitions.

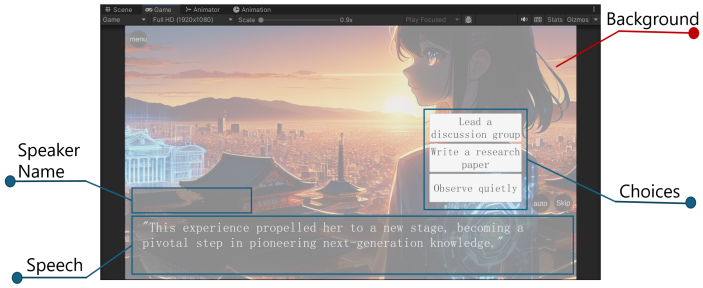


Fig. 1. The UI components of our Visual novel implementation in Unity, featuring standard elements such as the dialogue box (Speech), speaker name display, background image, and interactive choice buttons. The text content (Blue line) is generated by GPT-4o, and the background (Red line) is generated by DALL-E 3.

By eliminating the need for scripting, our system enables authors and designers to rapidly prototype or experience interactive fiction (Figure 1) by simply uploading a story file. The inserted choices are not scripted in advance but instead synthesized at runtime based on story context, as shown in Figure 2. This allows for emergent interactivity and re-playability, bridging the gap between linear fiction and interactive narrative design.

A full demonstration of the visual novel interface and generated backgrounds can be viewed on our static showcase website[1] and source code[2] is made publicly available.

II. Related Work

Traditional VN engines such as Ren'Py or TyranoScript allow authors to define scenes and branching logic via domain-specific scripting languages. While flexible, these systems require manual authoring of each choice and dialogue branch, limiting scalability. Camingue et al. [1] identified the lack of a unified definition for visual novels and proposed one by analyzing 30 prior academic definitions alongside 54 existing VN titles, highlighting four core design elements: art and aesthetics, narrative structure, interactivity, and story quality.

[1]Demo: https://zifanye.github.io/DemoOfVNstatic/
[2]Source code: https://github.com/ZifanYE/UnityVNforGCCE

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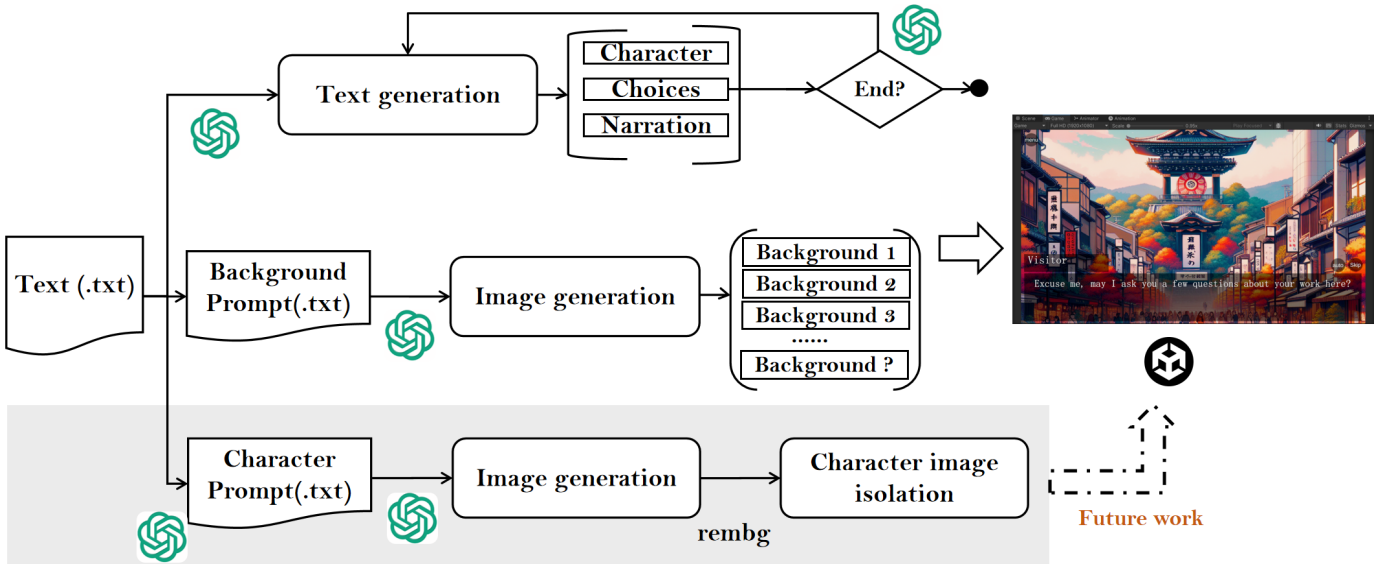


Fig. 2. The architecture of the whole system. Given an input story, the system synchronously generates segmented narrative text with interactive choices. Scene prompts are extracted and used to generate background images in order. Character images are pre-generated based on extracted character metadata and loaded when matched during dialogue.

Previous studies have demonstrated the capability of LLMs to generate characters, events, and branching options for visual novels. Structured Plot Graphs (SPGs) [2] showed knowledge graphs can help maintain logical consistency in story generation while also identifying key events to enhance narrative engagement. The Multiverse framework [3] introduced a dynamic approach, Dynamic Context Prompting/Programming (DCP/P), that allows for greater flexibility in generating branching narratives and improves both coherence and richness of the story. Gursesli et al. [4] confirmed keyword-guided generation can influence narrative direction effectively.

However, neither Gursesli et al. nor SPGs generates new text in response to player interactions. In addition, the Multiverse framework lacks integration with a real-time rendering presentation in a VN engine. In contrast, our system aims to automate the entire visual novel pipeline, with multi-turn generation that continuously pushes the story forward. It even allows the AI to create alternative outcomes, such as brand new bad endings, that are not presented in the source material. Table I summarizes the differences between the previous systems and ours.

Prior studies have addressed automatic dialogue generation and structured storytelling using language models. NarrativeGenie [5] introduced an LLM-based system capable of generating narrative beats and dynamic NPC conversations. HTEKG [6] proposed a human-trait-enhanced literary knowledge graph that incorporates character traits, events, and narrative roles to inform structured story generation.

In terms of character image preprocessing, rembg, based on the U2-Net [7] architecture, has been widely adopted for background removal. Additionally, BiRefNet [8] was developed for high-resolution binary segmentation tasks and has demonstrated effectiveness in precise character cutout applications.

Also, our work is related to another 2025 GCCE demo paper by Zhang et al. [9], which utilizes the Archaeological Prototyping (AP) model [10] to generate a story. The AP model constructs a generational sociocultural model using six entities and twelve directed relations, serving as a reference for an LLM to answer questions such as "What consequences might this lead to?" with the ultimate goal of story generation, thereby raising awareness and fostering critical thinking in the user. While the system of Zhang et al. [9] primarily aims to predict the future of a specific target, our system focuses on interactive storytelling within the visual novel framework, utilizing a given story generated by their system.

TABLE I
COMPARISON OF RELATED WORK AND OUR PROPOSED SYSTEM

System	LLM	Interactive	Visual novel	Dynamic
SPGs [2]	GPT-3.5	x	x	✓
Multiverse [3]	Claude 3 Opus	✓	web UI	✓
Chronicles [4]	ChatGPT (Web)	✓	not implemented	x
Ours	GPT-4o (OpenAI)	✓	✓	✓

III. PROPOSED SYSTEM

We present a VN framework, where the system is implemented in Unity and intended as a supplementary feature to illustrate the potential of LLM-guided branching narratives in an interactive medium. Our system takes as its input a text-based story generated from a system like that of Zhang et al. [9] and automatically segments it into scene-based episodes, each followed by multiple choice options to guide the branching narrative.

After starting the game, the interface displays a new visual scene, as shown in Figure 1. The system consists of several key

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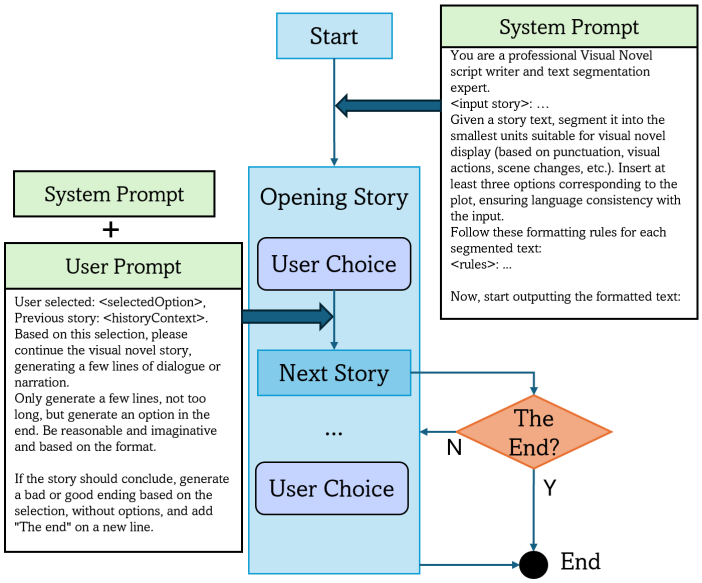


Fig. 3. The initial VN story is generated using a system prompt. After each choice, a user prompt is added with story history and branching intent to guide continuation toward a possible ending. If "The end" is generated, the runtime mode in Unity will quit.

UI components: a speech window for narration or character dialogue, a speaker name box indicating name or the identity of the current speaker, a dynamically rendered background scene, and overlay options for player decisions. Players can progress through the story by selecting one of the presented options, triggering real-time continuation of the narrative based on their choice.

A. Prompt Design

To support multi-turn generation, we design a prompt structure consisting of a role definition and format constraints, as shown in Figure 3. The system prompt is for the initial stage, including generating choice content. Every time the user selects the choice, it will trigger the next story generation. The next story generation can generate multiple endings to end the story before the user passes all scenarios from the original story. The new generated content will be inserted into the initial story generation.

System Prompt is used at the beginning of the story to generate an initial multi-paragraph segment.

User Prompt is used after each choice to continue the narrative based on previous context and the player-selected option.

For background generation, the following two tasks are designed:

Background Segmentation Prompt: Automatically segmenting the narrative into discrete background scenes and extracting descriptive cues for each.

Backgrounds Generation Prompt: Feeding these descriptive cues into DALL-E 3 to generate background images aligned with the narrative tone.

For character generation, prompts are similar to the ones for the background generation. We utilize the rembg tool to process generated images and remove backgrounds for transparent integration.

B. Story Input and Text Segmentation

As the input in this work, we used a story generated from Zhang et al. [9], which depicts a speculative 2050 scenario at Ritsumeikan University involving debates on AI ethics. This long-form story serves as the narrative foundation for our runtime generation system.

The system prompts the LLM in use, GPT-4o in this work, to segment the story into smaller narrative scenes based on structure, pacing, and semantic transitions. For instance, when the character Aoi is involved in a scene with other characters, the model may probabilistically insert dialogue lines such as:

> "–Aoi hesitated for a moment, then walked over to the professor with determination.–:"
> "–Professor, may I have a word with you?–:"
> "–Of course, Aoi. What concerns you about AI and education?–:"
> "–Aoi took a deep breath, organizing her thoughts.–:"

This process enables dynamic assignment of speaker roles and the emergence of natural multi-character conversations.

Segmented scenes are then passed into Unity along with generated visual assets and rendered as sequential visual novel episodes.

C. Choice Generation and Narrative Branching

Following each segmented scene, the system automatically generates two to three player choices that are contextually relevant to the preceding narrative. These choices are displayed in Unity as interactive options. Once the player selects one, the system will send the updated context back to GPT-4o, which continues the story in real time.

Stories generated from Zhang et al. [9] do not contain any branches or alternative endings, with varying lengths from approminately 300 to 2000 words. Each of them is treated as the default "true end", a convention commonly seen in visual novels where the base story concluded with a favorable or canonical resolution. To enhance interactivity and tension, our system is designed to occasionally insert early "bad endings" based on player decisions before the canonical conclusion is reached.

When GPT-4o determines that a branch should result in an early ending, it appends a terminal sentence as follows:

> "The end."

This will signal the termination of that storyline path and end the VN session.

An example of this is shown in Figure 4, where the original ending involves the protagonist Aoi overcoming a major ethical dilemma. However, due to the player's earlier decision, the system introduces a premature bad ending in which Aoi fails to resolve the conflict.

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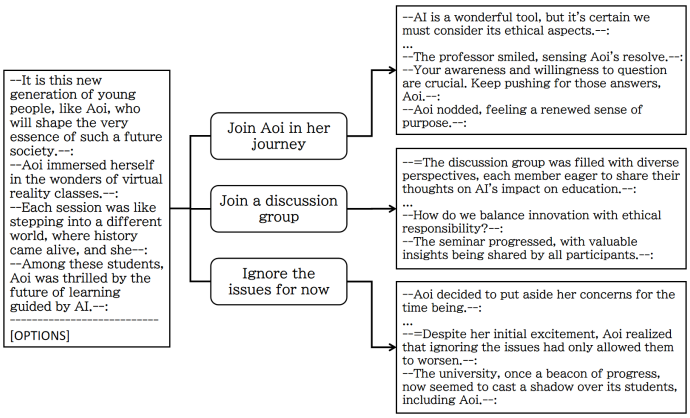


Fig. 4. Example of an inserted choice leading to endings based on the original story from [9], the third choice "Ignore the issues for now" would cause one bad ending.

Currently, our system generates both text and visual assets in real time and does not yet support saving, rollback, or branching persistence. As future work, our goal is to implement a rollback function that allows players to revisit previous decision nodes and regenerate alternative outcomes from earlier points. We also plan to add a branching memory mechanism that tracks past breakthroughs, enabling new paths to emerge atop previously explored stories.

Finally, while the system generally maintains the required formatting conventions, occasional generation errors (e.g. missing speaker labels or misaligned dialogues) may occur. We plan to incorporate format validation and fallback logic in future iterations to reduce these errors and provide a smoother narrative experience.

D. Multimodal Asset Generation

The background and character image generation prompts mentioned above are constructed based on scene-level descriptions and character attributes, as detailed in Section III-A, and the scene segmentation process, as illustrated in Figure 2. The generated backgrounds can be viewed on our static showcase website[2].

In our current implementation, while the system dynamically generates story segments and choices, it does not yet support real-time generation of character visuals that are fully coherent with the evolving narrative, as shown in Figure 2. To maintain consistency, we predefine a set of named characters. During game play, the system detects the speaker's name from each dialogue segment, and if it matches a predefined character name, it automatically retrieves and displays the corresponding character portrait. This approach ensures stable visual representation, even though the characters' appearances are not generated dynamically alongside the story.

IV. Conclusion and Future Work

In this study, we present an end-to-end system for automatically generating and deploying interactive visual novels from LLM-generated narrative text. By leveraging GPT-4o for scene segmentation, multi-turn continuation, and choice generation, combined with DALL-E 3 and rembg for multimodal asset synthesis, our framework enables the rapid creation of immersive, branching storytelling experiences in Unity. The system highlights how narrative logic, aesthetic components, and user interaction can be integrated with minimal manual intervention.

While the current implementation successfully automates most of the visual novel pipeline, several limitations remain. The system does not yet perform dynamic scene generation for newly generated story branches, nor does it incorporate feedback from player interactions to control the timing or appropriateness of scene transitions. In addition, formatting errors occasionally emerge during generation due to the complexity of enforcing narrative conventions in freeform text.

As future work, we plan to enhance runtime control over visual elements, allowing the system to decide when to introduce new backgrounds and characters based on narrative flow. We also aim to support dynamic background generation for new story branches rather than relying on static prompts. Furthermore, we will implement a rollback mechanism, enabling players to revisit previous nodes and re-generate alternate story paths. These improvements will further advance narrative coherence, interactivity, and replayability in AI-driven storytelling systems.

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